NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

FAST PESHAWAR CAMPUS

ASSIGNMENT: 2

GROUP MEMBERS: IFFA USMAN QAZI (24P-6700)

FARDA SHAHID (24P-3013)

COURSE: SOFTWARE QUALITY ENGINEERING

COURSE CODE: SE3002 SE-5A

SUBMITTED TO: MA'AM SARA REHMAT

PROGRAM: SOFTWARE ENGINEERING (SE)

SEMESTER: 5th

DATE: 11th NOVEMBER, 2026



Testing Report

1. Black-box Test Design:

Expected results were derived from the SRS (REQUIREMENTS.md, FR-01 to FR-15) before the implementation was inspected. The suite is in tests/test_blackbox.py; the design sheets are docs/blackbox_test_design.xlsx and docs/decision_table.xlsx.

Summary:

Three requirement-based techniques produced 47 designed cases plus 11 state-change tests (58 test runs). ECP partitions the inputs of each rule (credit hours, dues, prerequisites, availability) into valid and invalid classes, with one representative per class. BVA tests the values at and just outside each limit: current credits 0 to 18, course credits 1 to 4, the combined 18-credit maximum, capacity and enrollment, and dues around zero. The decision table encodes the five-condition priority of registration_decision() as six rules plus four priority-conflict combinations. Optimization found three groups of exact duplicates: 8 tests collapse to 3 executions, so 58 test runs become 53 unique executions while every technique ID stays traceable.

Technique	**Test IDs**	**Count**
Equivalence Class Partitioning	ECP01 to ECP17	17
Boundary Value Analysis	BVA01 to BVA20	20
Decision Table	DT01 to DT10	10
State-change tests	ST01 to ST08 (ST03a to ST03d)	11
Total		**58**

Detailed Blackbox Tests available in balckbox_test_design.xlsx file.

1.1 Equivalence Class Partitioning (ECP):

Test ID	**Req**	**Class**	**Input**	**Expected**
ECP01	FR-04	Dues = 0 (valid)	dues 0	REGISTERED
ECP02	FR-04	Dues > 0	dues 500	DUES_BLOCK
ECP03	FR-15	Dues < 0 (invalid)	dues -1	ValueError "Invalid dues"

Test ID	Req	Class	Input	Expected
ECP06	FR-06	Some prerequisites missing	required {CS101, MATH101}, done {CS101}	PREREQUISITES_MISSING
ECP08	FR-05	Already registered	CS201 already registered	ALREADY_REGISTERED
ECP10	FR-08	Course full	enrolled 30, capacity 30	WAITLISTED
ECP11	FR-02	Invalid course credits	check_credit_limit(10, 5)	ValueError "Invalid course credits"
ECP12	FR-15	Invalid capacity	capacity 0	ValueError "Invalid capacity or enrollment"

1.2 Boundary Value Analysis (BVA):

Test ID	Boundary	Input	Expected
BVA01	Current credits: min - 1	(-1, 3)	ValueError (current credits)
BVA02	Current credits: min	(0, 3)	True
BVA04	Combined total exactly 18	(17, 1)	True
BVA05	Combined total 19	(18, 1)	False
BVA06	Current credits: max + 1	(19, 1)	ValueError (current credits)
BVA07	Course credits: min - 1	(5, 0)	ValueError (course credits)
BVA10	Course credits: max + 1	(5, 5)	ValueError (course credits)
BVA13	Enrolled = capacity - 1	29 / 30	REGISTERED
BVA14	Enrolled = capacity	30 / 30	WAITLISTED
BVA15	Dues = 0	0	REGISTERED
BVA16	Dues = 1	1	DUES_BLOCK
BVA17	Enrolled = capacity + 1	31 / 30	ValueError (capacity or enrollment)



1.3 Decision Table:

Priority (FR-09): dues, already registered, prerequisites, credit limit, seat availability. "-" means the condition is not evaluated because an earlier rule already decided the outcome.

Priority-combination tests: DT07 (all conditions true, dues wins), DT08 (already registered beats missing prerequisites), DT09 (missing prerequisites beat credit limit and full course), DT10 (credit limit beats full course).

Detailed Decision Table available in decision_table.xlsx file.

Condition / Rule	**R1**	**R2**	**R3**	**R4**	**R5**	**R6**
Dues > 0	T	F	F	F	F	F
Already registered	-	T	F	F	F	F
Prerequisites missing	-	-	T	F	F	F
Credit total > 18	-	-	-	T	F	F
Course full	-	-	-	-	T	F
Outcome	DUES_BLOCK	ALREADY_REGISTERED	PREREQUISITES_MISSING	CREDIT_LIMIT	WAITLISTED	REGISTERED
Test	DT01	DT02	DT03	DT04	DT05	DT06

Expected values come only from the SRS. Where a test failed, the expected value was not changed.

1.4 Optimization (A5):

Tests were compared on inputs, state, expected output and purpose. Identical outputs alone were not treated as duplication.

Exact duplicates: ECP10, BVA14 and DT05 use the same input (enrolled 30 of 30 for an eligible student) and expect WAITLISTED. ECP01 and BVA15 also share the same input (dues 0).



Not duplicates: ECP02 (dues 500), BVA16 (dues 1) and DT01 (dues 100) use different inputs and test different things. BVA12 (16 + 3) and DT04 (17 + 3) also differ.

Execution can be consolidated while keeping every technique ID mapped to its test in the Traceability sheet of blackbox_test_design.xlsx.

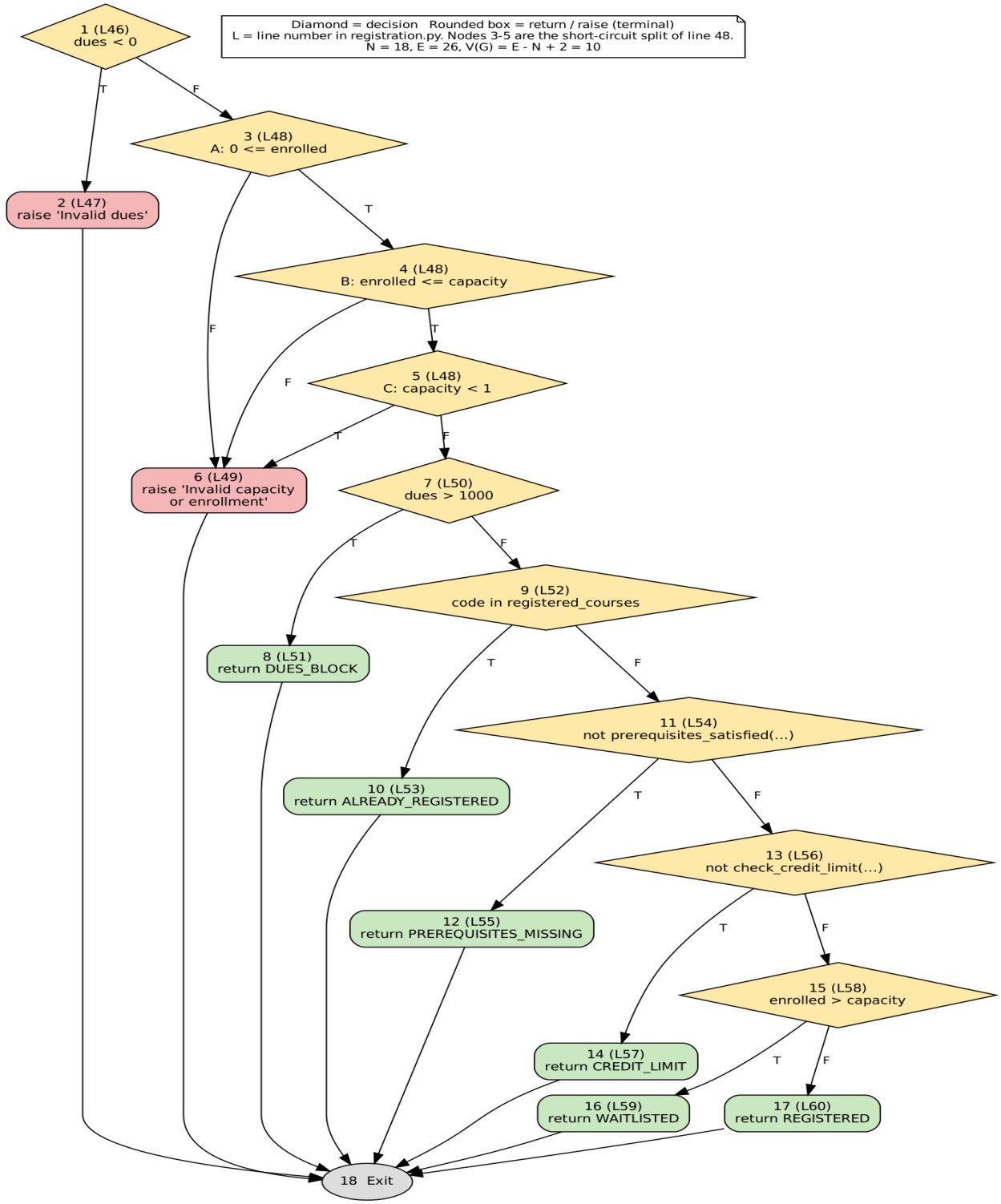
1.5 Initial execution result:

The first run of tests/test_blackbox.py gave 16 failed, 42 passed. The failures were preserved in docs/blackbox_initial_run.txt and the failing GitHub Actions run, and expected values were not changed.

2. White-box Analysis:

The tests are in tests/test_whitebox.py. Each test documents the branch, condition or path it targets.

2.1 Control Flow Graph:



Node	Type	Statement
N1	Decision	dues < 0
N2	Raise	"Invalid dues"
N3	Decision	A: 0 <= enrolled
N4	Decision	B: enrolled <= capacity
N5	Decision	C: capacity < 1
N6	Raise	"Invalid capacity or enrollment"
N7	Decision	dues > 1000 (original code)
N8	Return	DUES_BLOCK
N9	Decision	course code in registered_courses
N10	Return	ALREADY_REGISTERED
N11	Decision	prerequisites not satisfied
N12	Return	PREREQUISITES_MISSING
N13	Decision	credit limit not OK
N14	Return	CREDIT_LIMIT
N15	Decision	enrolled > capacity (original code)
N16	Return	WAITLISTED
N17	Return	REGISTERED
N18	Exit	exit

2.2 Cyclomatic Complexity:

- Nodes N = 18, edges E = 26.
- **V(G) = E - N + 2 = 26 - 18 + 2 = 10.**
- **Decision nodes** = **9 + 1 = 10**.
- Hence, both the results match.



2.3 Independent Basis Paths:

Path	Nodes	Outcome	Test
P01	1-2-18	Invalid dues	WB-P01
P02	1-3-6-18	enrolled < 0	WB-P02
P03	1-3-4-6-18	enrolled > capacity	WB-P03
P04	1-3-4-5-6-18	capacity < 1	WB-P04
P05	1-3-4-5-7-8-18	DUES_BLOCK	WB-P05
P06	1-3-4-5-7-9-10-18	ALREADY_REGISTERED	WB-P06
P07	1-3-4-5-7-9-11-12-18	PREREQUISITES_MISSING	WB-P07
P08	1-3-4-5-7-9-11-13-14-18	CREDIT_LIMIT	WB-P08
P09	1-3-4-5-7-9-11-13-15-16-18	WAITLISTED	WB-P09
P10	1-3-4-5-7-9-11-13-15-17-18	REGISTERED	WB-P10

2.5 MC/DC Analysis:

Decision on line 48: not (A and B) or C, where A: 0 <= enrolled, B: enrolled <= capacity, C: capacity < 1. Python short-circuits: B is evaluated only if A is True, and C only if not (A and B) is False. Result "NE" means not evaluated.

Test	A	B	C	Decision	Pair shows
WB-14	T	T	F	False	base
WB-15	F	NE	NE	True	WB-14/15: effect of A
WB-16	T	F	NE	True	WB-14/16: effect of B
WB-17	T	T	T	True	WB-14/17: effect of C

Four tests (N + 1 for three conditions) demonstrate the independent effect of each condition. Calculated manually, since pytest-cov does not measure MC/DC.

Decision on line 70 (register_course): decision == "WAITLISTED" and student_id not in waitlist, with P the first condition and Q the second.

Test	P	Q	Decision	Pair shows
WB-11	T	T	True	base
WB-12	T	F	False	WB-11/12: effect of Q
WB-13	F	NE	False	WB-11/13: effect of P

2.6 Coverage Comparison (statement + branch, pytest-cov):

Stage	Tests	Result	Coverage	Missing lines
Initial (black-box only)	58	16 failed, 42 passed	93%	51, 59, 71
After white-box tests, before fixes	75	19 failed, 56 passed	95%	59, 71
After defect fixes	75	75 passed	100%	none

Gaps and added tests

- Line 51 (DUES_BLOCK return) was not reached by black-box tests, which used dues of 500 or less. WB-P05 covers it with dues of 5000.
- Lines 59 and 71 (WAITLISTED return and waitlist append) were unreachable until defect 4 was fixed. WB-P09, WB-11 and WB-12 target them.
- Added tests: WB-P01 to WB-P10 and WB-11 to WB-17.

Evidence files: docs/coverage_initial.txt, docs/coverage_final_before_fix.txt, docs/coverage_final_after_fix.txt, docs/final_run.txt.

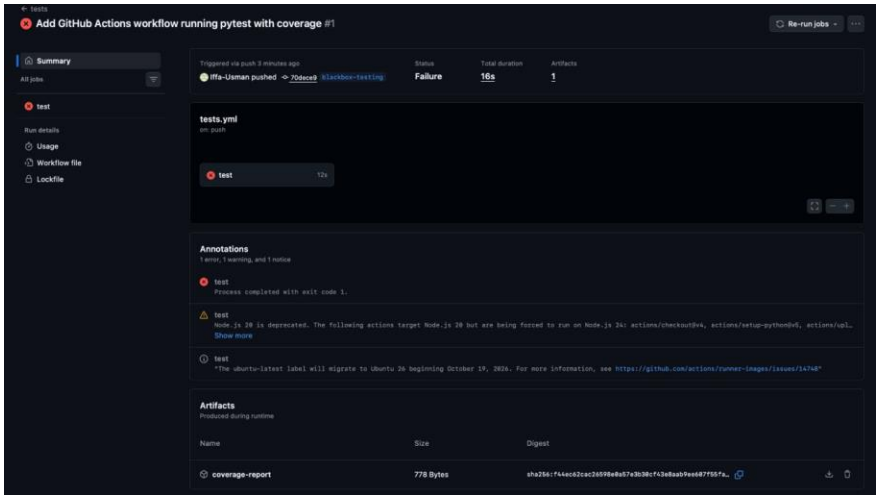
3. Defects:

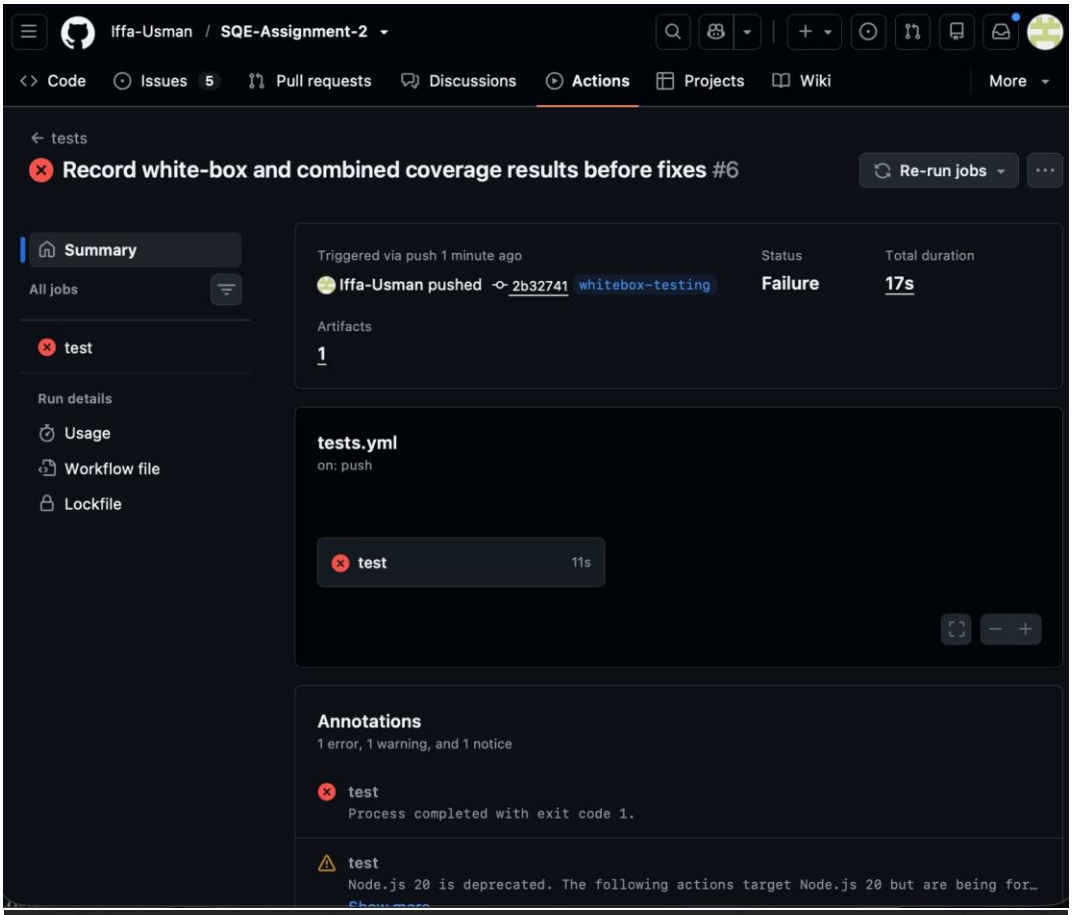
The 19 failing tests (16 black-box, 3 white-box) trace to five distinct defects in registration.py. Each has one GitHub Issue.

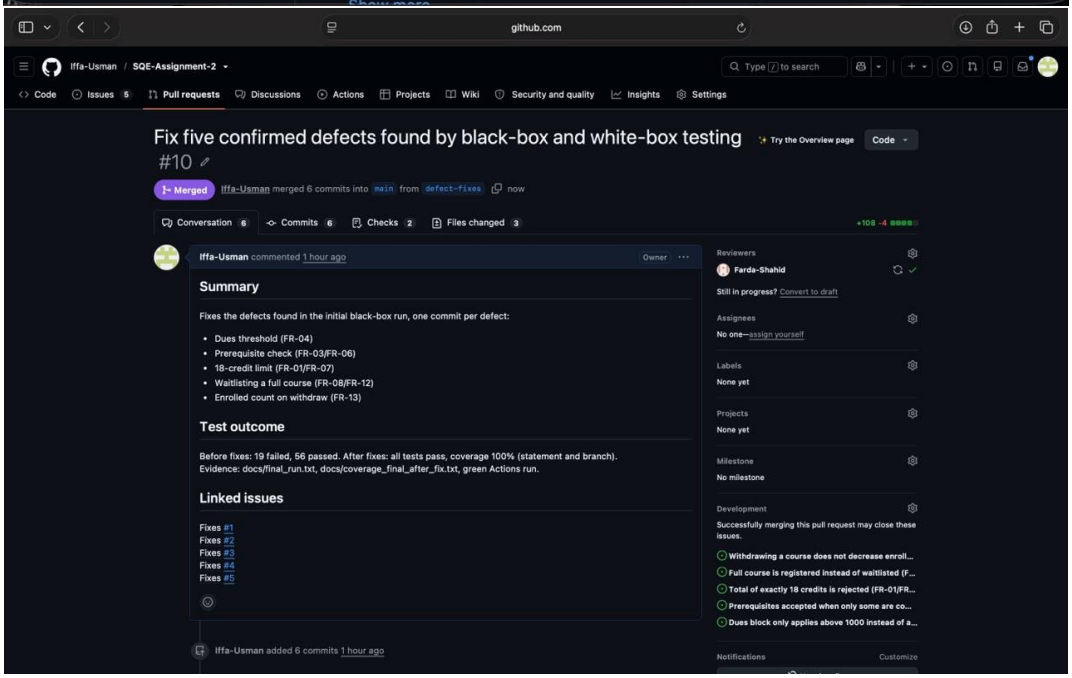
#	Defect	Req	Failing tests	Issue	Status
1	Dues block applies only above 1000 instead of any dues above 0	FR-04	ECP02, BVA16, DT01, DT07, ST03a	#1	Fixed
2	Prerequisites accepted when only some are completed	FR-03, FR-06	ECP06, ECP14	#2	Fixed
3	Total of exactly 18 credits rejected	FR-01, FR-07	BVA04, BVA11, ST08	#3	Fixed
4	Full course registered instead of waitlisted	FR-08, FR-12	ECP10, BVA14, DT05, ST02, WB-P09, WB-11, WB-12	#4	Fixed
5	Withdraw does not decrement the enrolled count	FR-13	ST04, ST07	#5	Fixed

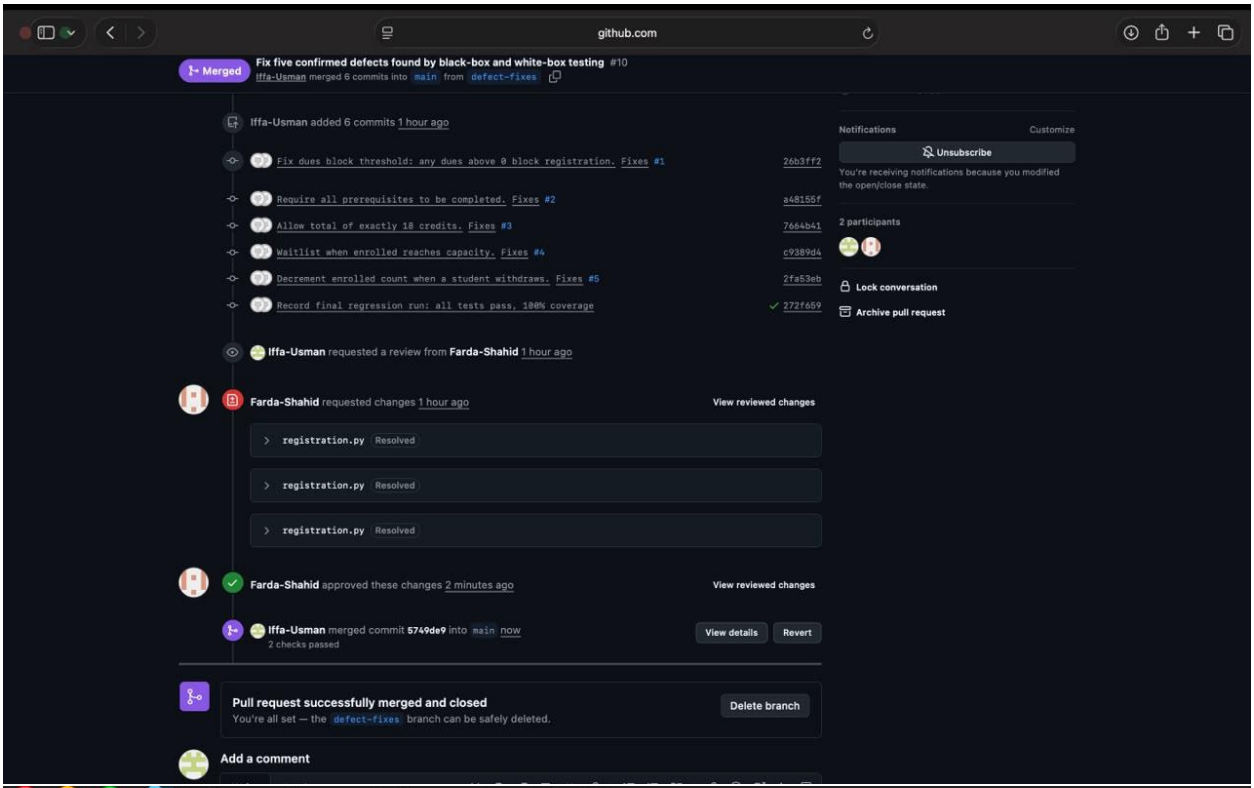
Fixes (branch defect-fixes, one commit per defect): dues threshold changed to 0; prerequisites check changed to a subset test; credit limit changed to allow 18; waitlist check changed to enrolled >= capacity; enrolled decremented on withdraw.

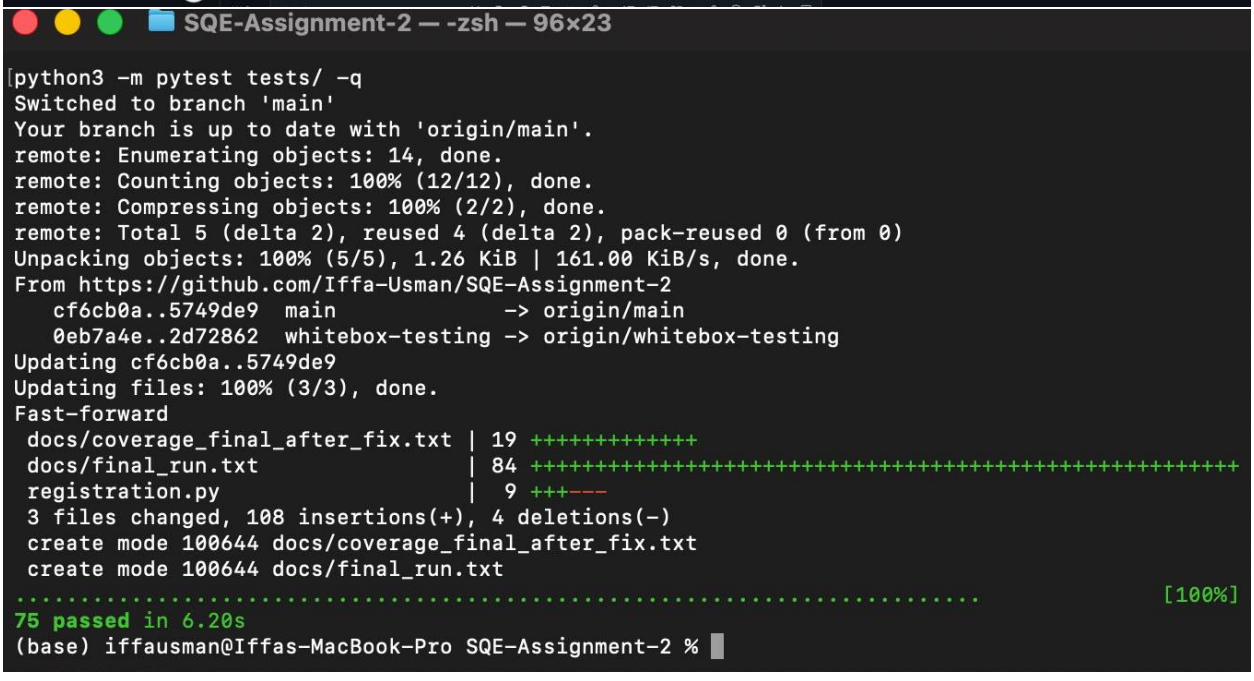
Evidence: the failing GitHub Actions run before the fixes (initial black-box run) and the passing run after the fixes are preserved as screenshots and in the repository's Actions history.




Iffa-Usman / SQE-Assignment-2
Code
Issues 5
Pull requests
Discussions
Actions
Projects
Wiki
More
tests
Record white-box and combined coverage results before fixes #6
Re-run jobs
Summary
All jobs
test
Run details
Usage
Workflow file
Lockfile
Triggered via push 1 minute ago
Iffa-Usman pushed 2b32741 whitebox-testing
Status
Failure
Total duration
17s
Artifacts
1
tests.yml
on: push
test 11s
Annotations
1 error, 1 warning, and 1 notice
test
Process completed with exit code 1.
test
Node.js 20 is deprecated. The following actions target Node.js 20 but are being for…


github.com
Iffa-Usman / SQE-Assignment-2
Code
Issues 5
Pull requests
Discussions
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Projects
Wiki
Security and quality
Insights
Settings
Fix five confirmed defects found by black-box and white-box testing #10
Try the Overview page
Code
Merged
Iffa-Usman merged 6 commits into main from defect-fixes now
Conversation 6
Commits 6
Checks 2
Files changed 3
Iffa-Usman commented 1 hour ago
Owner
Summary
Fixes the defects found in the initial black-box run, one commit per defect:
Dues threshold (FR-04)
Prerequisite check (FR-03/FR-06)
18-credit limit (FR-01/FR-07)
Waitlisting a full course (FR-08/FR-12)
Enrolled count on withdraw (FR-13)
Test outcome
Before fixes: 19 failed, 56 passed. After fixes: all tests pass, coverage 100% (statement and branch).
Evidence: docs/final_run.txt, docs/coverage_final_after_fix.txt, green Actions run.
Linked issues
Fixes #1
Fixes #2
Fixes #3
Fixes #4
Fixes #5
Iffa-Usman added 6 commits 1 hour ago
Reviewers
Farda-Shahid
Still in progress? Convert to draft
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
Withdrawing a course does not decrease enroll...
Full course is registered instead of waitlisted (F...
Total of exactly 18 credits is rejected (FR-01/FR...
Prerequisites accepted when only some are co...
Dues block only applies above 1000 instead of a...
Notifications
Customize




SQE-Assignment-2 — -zsh — 96×23
python3 -m pytest tests/ -q
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
remote: Enumerating objects: 14, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 5 (delta 2), reused 4 (delta 2), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.26 KiB | 161.00 KiB/s, done.
From https://github.com/Iffa-Usman/SQE-Assignment-2
cf6cb0a..5749de9 main -> origin/main
0eb7a4e..2d72862 whitebox-testing -> origin/whitebox-testing
Updating cf6cb0a..5749de9
Updating files: 100% (3/3), done.
Fast-forward
docs/coverage_final_after_fix.txt | 19 ++++++++++++
docs/final_run.txt | 84 ++++++++++++

4. GitHub Repository, Pull Requests and Contributions:

Source: https://github.com/Iffa-Usman/SQE-Assignment-2

PR	Branch	Purpose	Author	Reviewer	Status	Link
#6	blackbox-testing	Black-box tests, design sheets, workflow	Iffa Usman	Farda Shahid	Approved, merged	PR #6
#7	whitebox-testing	White-box tests, CFG	Farda Shahid	Iffa Usman	Approved, merged	PR #7
#8	blackbox-testing	Review feedback applied to black-box tests (changes requested in PR #6)	Iffa Usman	Reviewed in PR #6	Merged	PR #8
#9	whitebox-testing	Review feedback applied to white-box tests (changes requested in PR #7)	Iffa Usman	Reviewed in PR #7	Merged	PR #9
#10	defect-fixes	Fixes for defects 1 to 5 (closes #1 to #5)	Iffa Usman	Farda Shahid	Approved, merged	PR #10

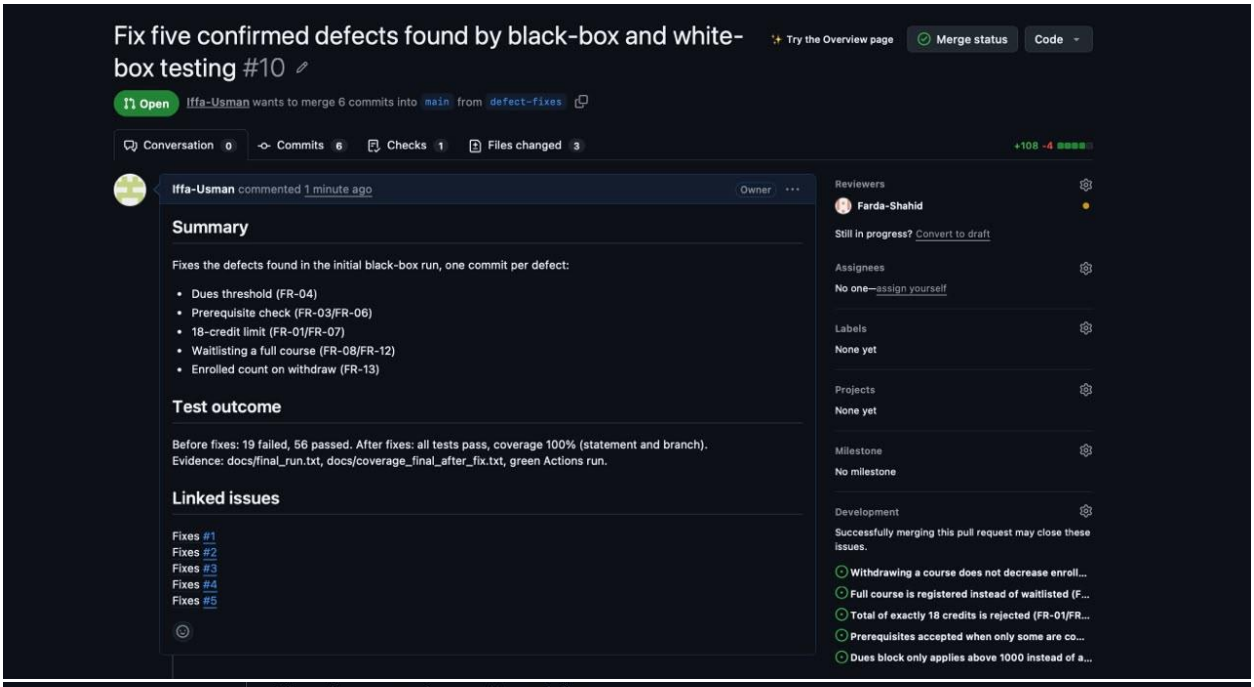
Review summary:
Each member reviewed the other member's pull request: Farda Shahid reviewed PR #6 (black-box) and PR #10 (defect fixes), and Iffa Usman reviewed PR #7 (white-box). Reviewers left line comments on the Files changed tab, authors replied, and the reviewers approved before merging. PR #8 and PR #9 only carried the review-feedback commits from the already-reviewed PR #6 and PR #7 branches into main, so they were merged without a second review. The GitHub Actions checks failed on the PRs that still contained the defects, and passed on PR #10 after the fixes.

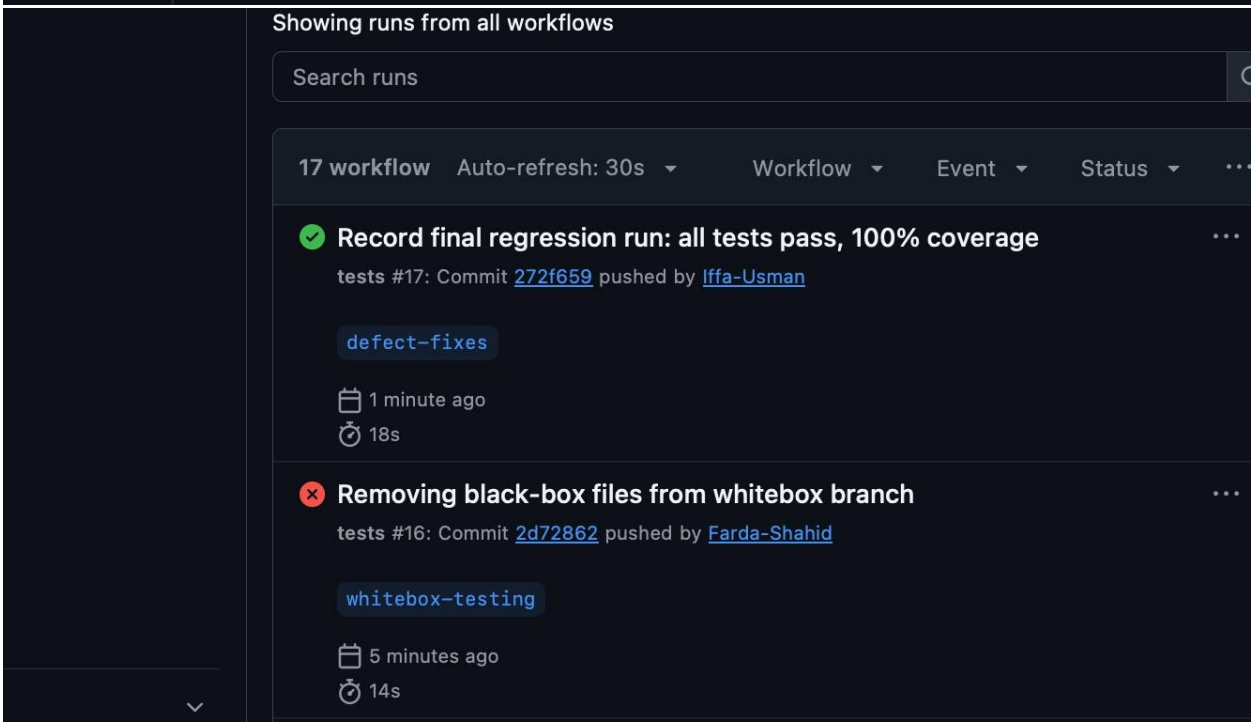


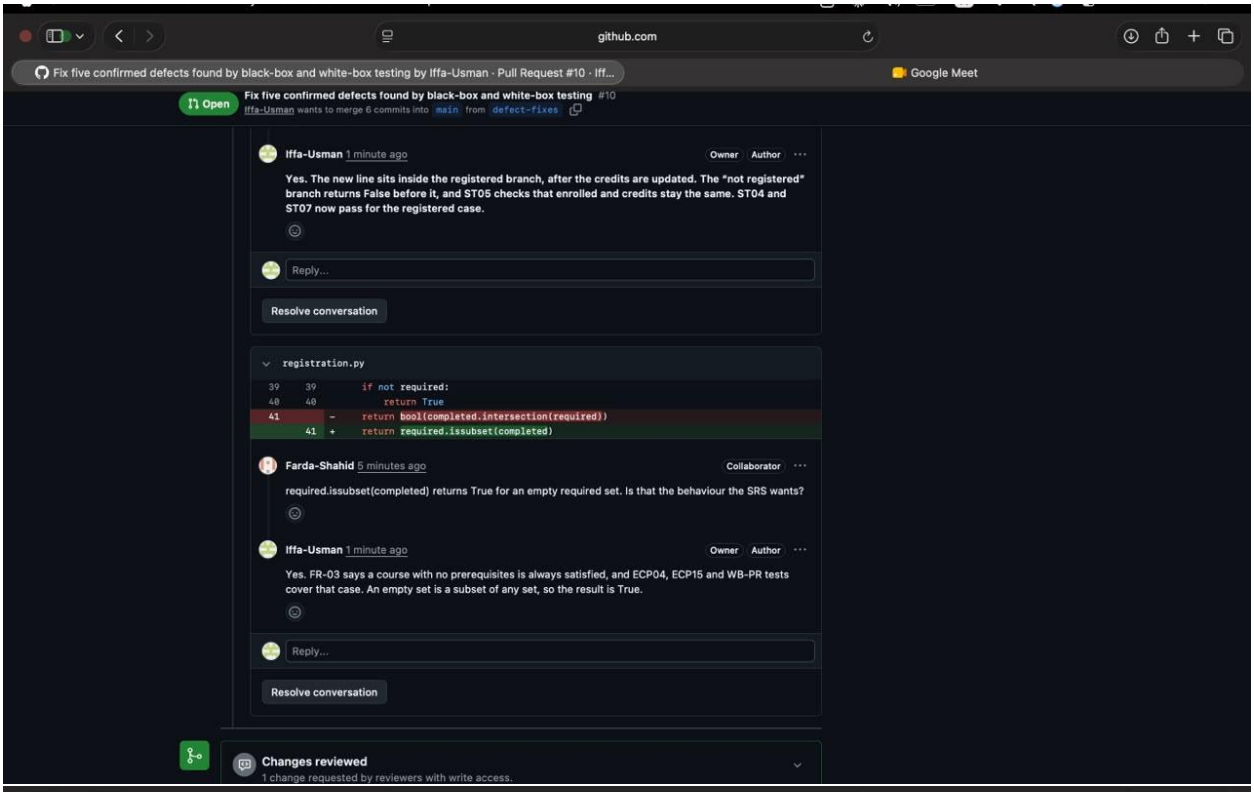
Individual contributions:

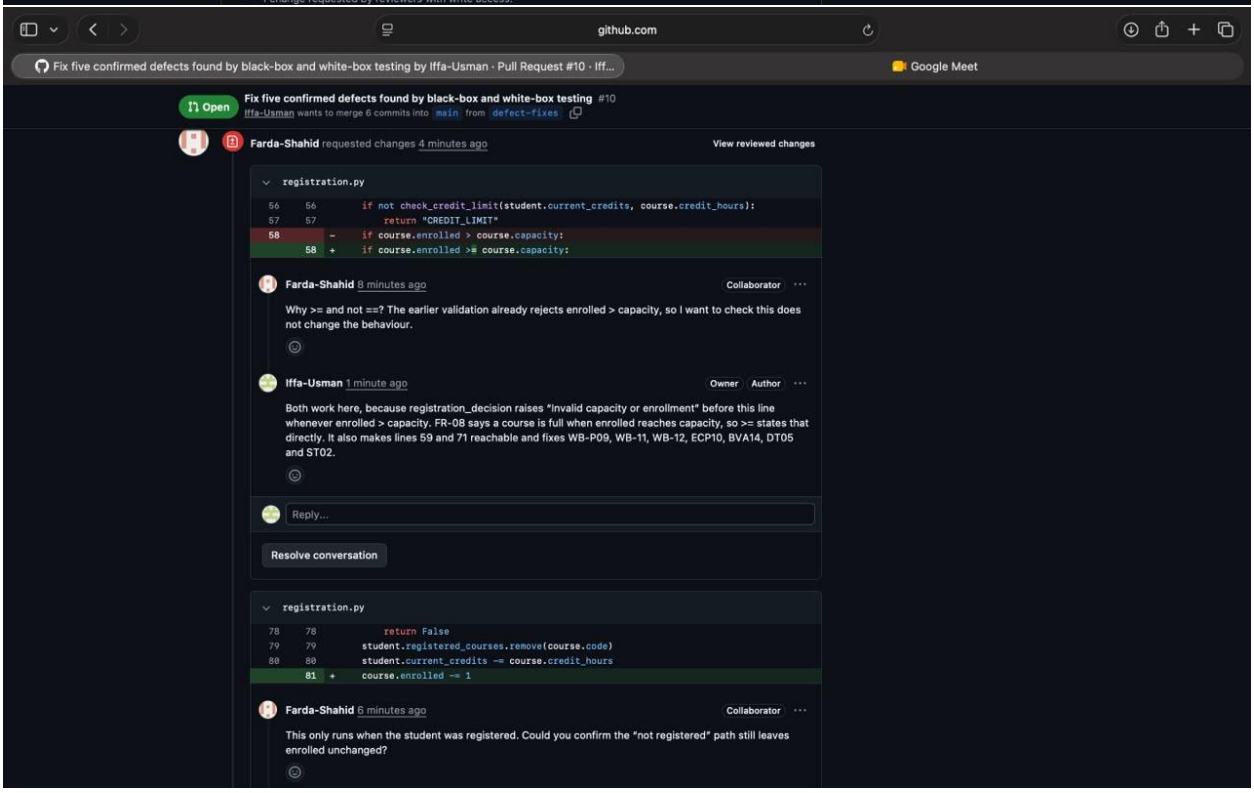
Member	**Contribution**
Iffa Usman	Created the repository; designed and implemented the black-box suite (58 tests: ECP, BVA, decision table, state changes); identified duplicate tests for optimization
Farda Shahid	Drew the CFG; calculated V(G) under two conventions; derived the 10 basis paths; measured initial and final coverage; performed the MC/DC analysis; implemented the white-box suite (17 tests)
Both	Report writing, pull request reviews, issue reporting and defect fixes

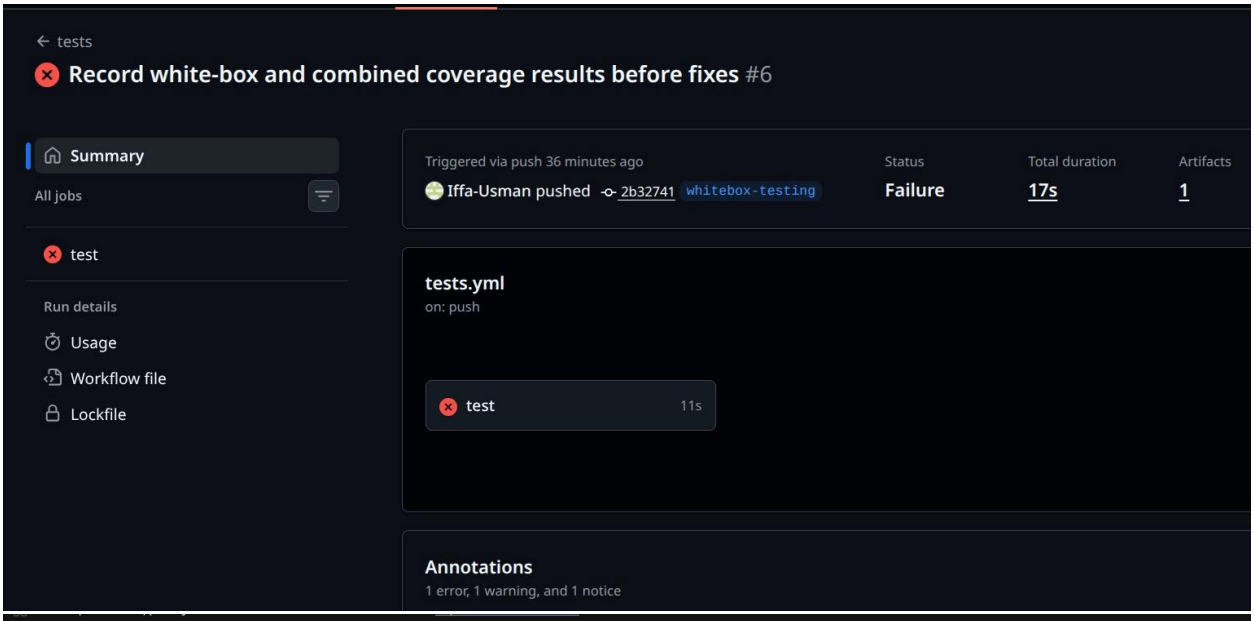
Evidence (Screenshots):

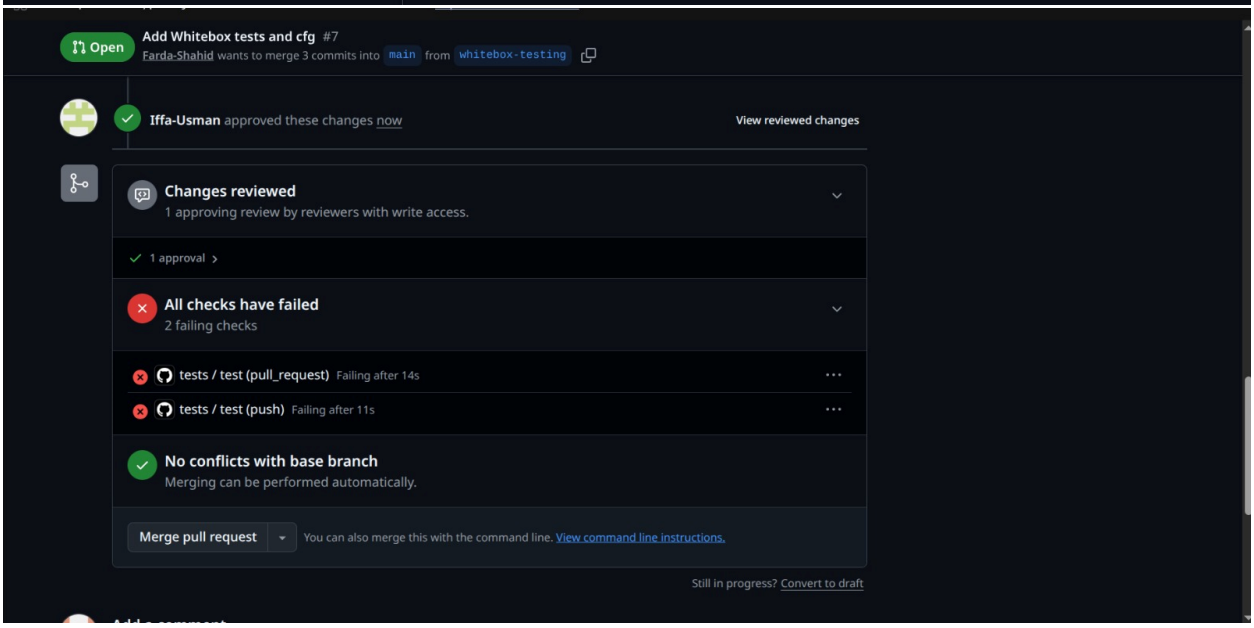





github.com
Fix five confirmed defects found by black-box and white-box testing by Iffa-Usman · Pull Request #10 · Iff...
Google Meet
Open
Fix five confirmed defects found by black-box and white-box testing #10
Iffa-Usman wants to merge 6 commits into main from defect-fixes
Iffa-Usman 1 minute ago
Owner Author
Yes. The new line sits inside the registered branch, after the credits are updated. The "not registered" branch returns False before it, and ST05 checks that enrolled and credits stay the same. ST04 and ST07 now pass for the registered case.
Reply...
Resolve conversation
registration.py
39 39 if not required:
40 40 return True
41 - return bool(completed.intersection(required))
41 + return required.issubset(completed)
Farda-Shahid 5 minutes ago
Collaborator
required.issubset(completed) returns True for an empty required set. Is that the behaviour the SRS wants?
Iffa-Usman 1 minute ago
Owner Author
Yes. FR-03 says a course with no prerequisites is always satisfied, and ECP04, ECP15 and WB-PR tests cover that case. An empty set is a subset of any set, so the result is True.
Reply...
Resolve conversation
Changes reviewed
1 change requested by reviewers with write access.


github.com
Fix five confirmed defects found by black-box and white-box testing by Iffa-Usman · Pull Request #10 · Iff...
Google Meet
Open
Fix five confirmed defects found by black-box and white-box testing #10
Iffa-Usman wants to merge 6 commits into main from defect-fixes
Farda-Shahid requested changes 4 minutes ago
View reviewed changes
registration.py
56 56 if not check_credit_limit(student.current_credits, course.credit_hours):
57 57 return "CREDIT_LIMIT"
58 - if course.enrolled > course.capacity:
58 + if course.enrolled >= course.capacity:
Farda-Shahid 8 minutes ago
Collaborator
Why >= and not ==? The earlier validation already rejects enrolled > capacity, so I want to check this does not change the behaviour.
Iffa-Usman 1 minute ago
Owner Author
Both work here, because registration_decision raises "invalid capacity or enrollment" before this line whenever enrolled > capacity. FR-08 says a course is full when enrolled reaches capacity, so >= states that directly. It also makes lines 59 and 71 reachable and fixes WB-P09, WB-11, WB-12, ECP10, BVA14, DT05 and ST02.
Reply...
Resolve conversation
registration.py
78 78 return False
79 79 student.registered_courses.remove(course.code)
80 80 student.current_credits -= course.credit_hours
81 + course.enrolled -= 1
Farda-Shahid 6 minutes ago
Collaborator
This only runs when the student was registered. Could you confirm the "not registered" path still leaves enrolled unchanged?


tests
Record white-box and combined coverage results before fixes #6
Summary
All jobs
test
Run details
Usage
Workflow file
Lockfile
Triggered via push 36 minutes ago
Iffa-Usman pushed 2b32741 whitebox-testing
Status
Failure
Total duration
17s
Artifacts
1
tests.yml
on: push
test 11s
Annotations
1 error, 1 warning, and 1 notice


Open
Add Whitebox tests and cfg #7
Farda-Shahid wants to merge 3 commits into main from whitebox-testing
Iffa-Usman approved these changes now
View reviewed changes
Changes reviewed
1 approving review by reviewers with write access.
1 approval
All checks have failed
2 failing checks
tests / test (pull_request) Failing after 14s
tests / test (push) Failing after 11s
No conflicts with base branch
Merging can be performed automatically.
Merge pull request
You can also merge this with the command line. View command line instructions.
Still in progress? Convert to draft